Method of expressing concentration of solution

41. (c) $M_1V_1 + M_2V_2 = M_3V_3;$

$1.5 \times 480 + 1.2 \times 520 = M \times 1000$

$M = \frac{720+624}{1000} = 1.344M.$

44. (c) $m = \frac{18\times1000}{180\times500} = 0.2m$

45. (a) Molarity $= \frac{\%\times10\times d}{GMM} = \frac{22\times10\times1.253}{342} = 0.805M.$

Normality $= \frac{\%\times10\times d}{GEM} = \frac{22\times10\times1.253}{342/6} = 4.83N$

Molality $= \frac{22\times1000}{342(100-22)} = 0.825m$

46. (a) 100 $ml.$ of 0.30M= $\frac{100\times0.3}{1000}$ = 0.03 mole of $NaCl$

100 ml of 0.40M = $\frac{100\times0.4}{1000}$ = 0.04 mole of $NaCl$

Moles of $NaCl$ to be added = 0.04 − 0.03 = 0.01mole

= 0.585 gm

47. (c) $N = \frac{6\times1000}{40\times100} = 1.5N$

It is show highest normality than others.

48. (b) $M = \frac{n}{V(l)} \Rightarrow 0.8 = \frac{0.1}{V(l)} \Rightarrow V = 125ml.$

50. (c) Strength of $H_2SO_4 = 98 \times 19.8g/litre$

$S = eq.wt. \times N;\ \ N = \frac{S}{eq.wt.} = \frac{98 \times 19.8}{49} = 39.6$

51. (c) $W = 1000gm\ \ (H_2O);\ n = 1\ mole$

$N = \frac{W}{M} = \frac{1000}{18} = 55.55$

$x_{\text{Solute}} = \frac{n}{n+N} = \frac{1}{1+55.55} = 0.018.$

53. (d) Normality of acid = molarity × basicity

i.e., 0.2=molarity × 2

∴ Molarity = 0.2/2 = 0.1

55. (b) Mole fraction of $H_2O = \frac{\frac{80}{18}}{\frac{80}{18}+\frac{20}{24}} = \frac{68}{77}.$

59. (d) Volume strength $= \frac{1.5 \times 100}{17} = 8.82.$

60. (a) $n = \frac{w}{m}; w = n \times m = 0.25 \times 98 = 24.5gm$